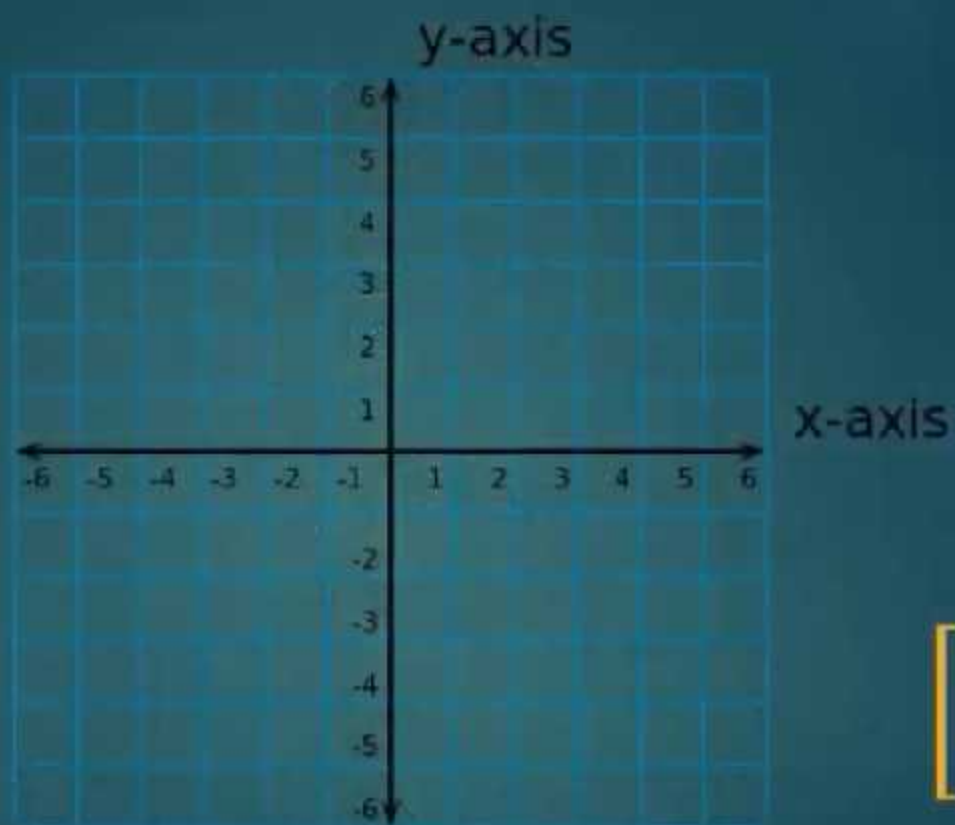
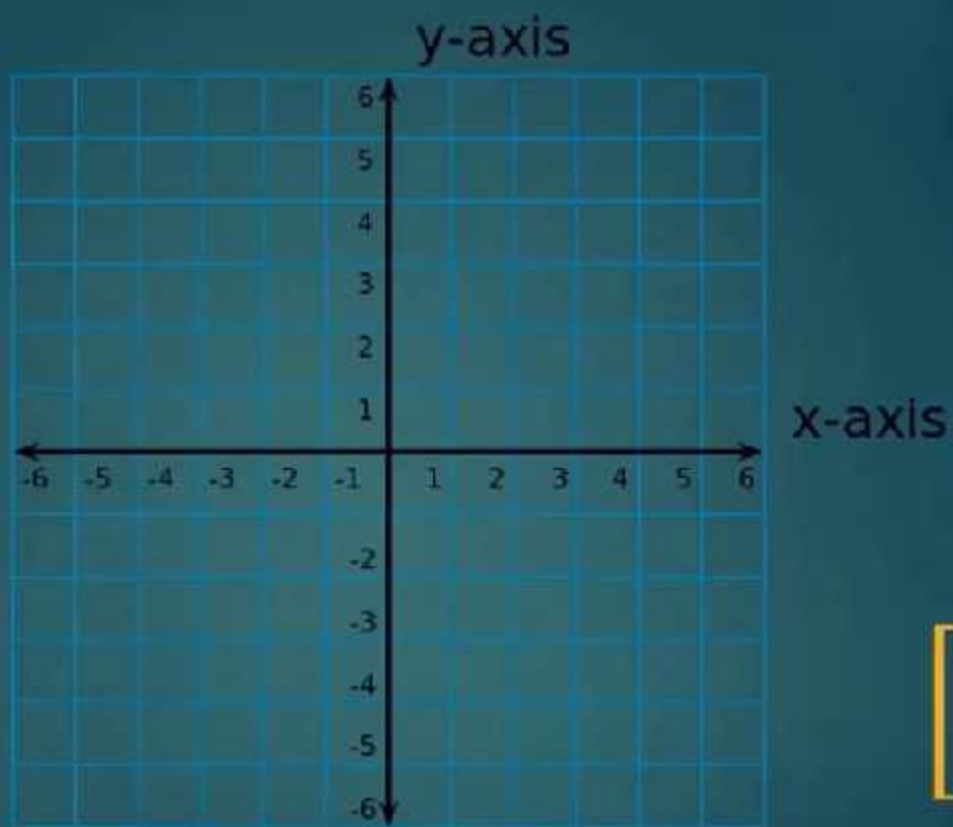


Vectors



Vectors

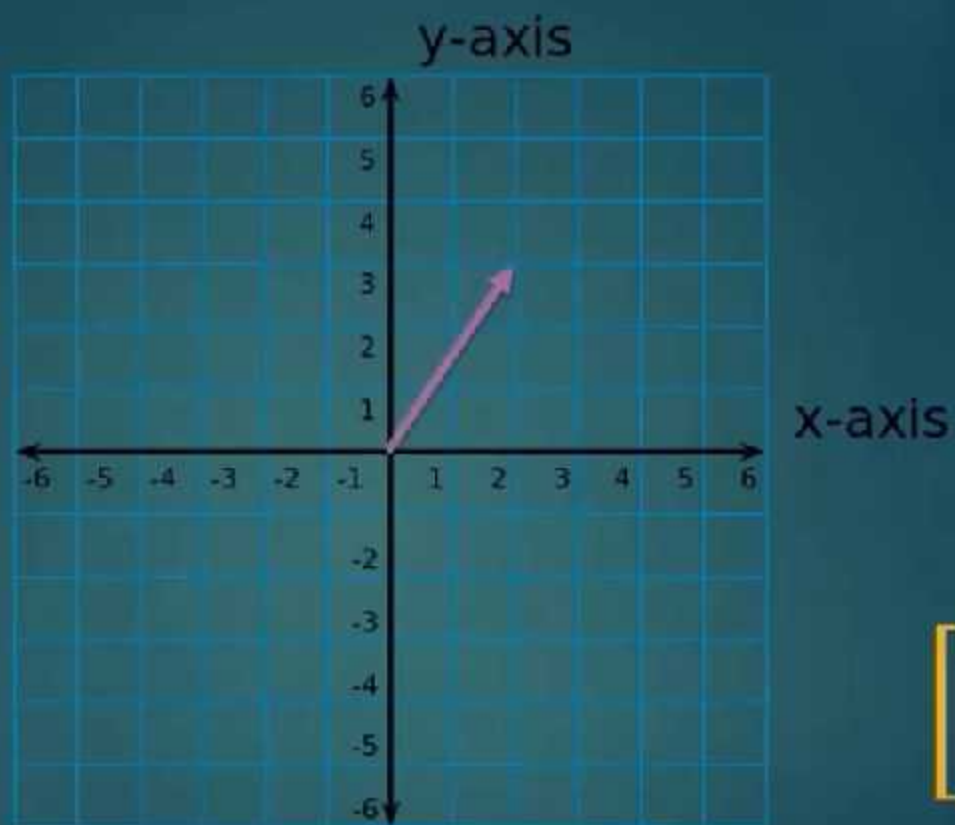
$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$



$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

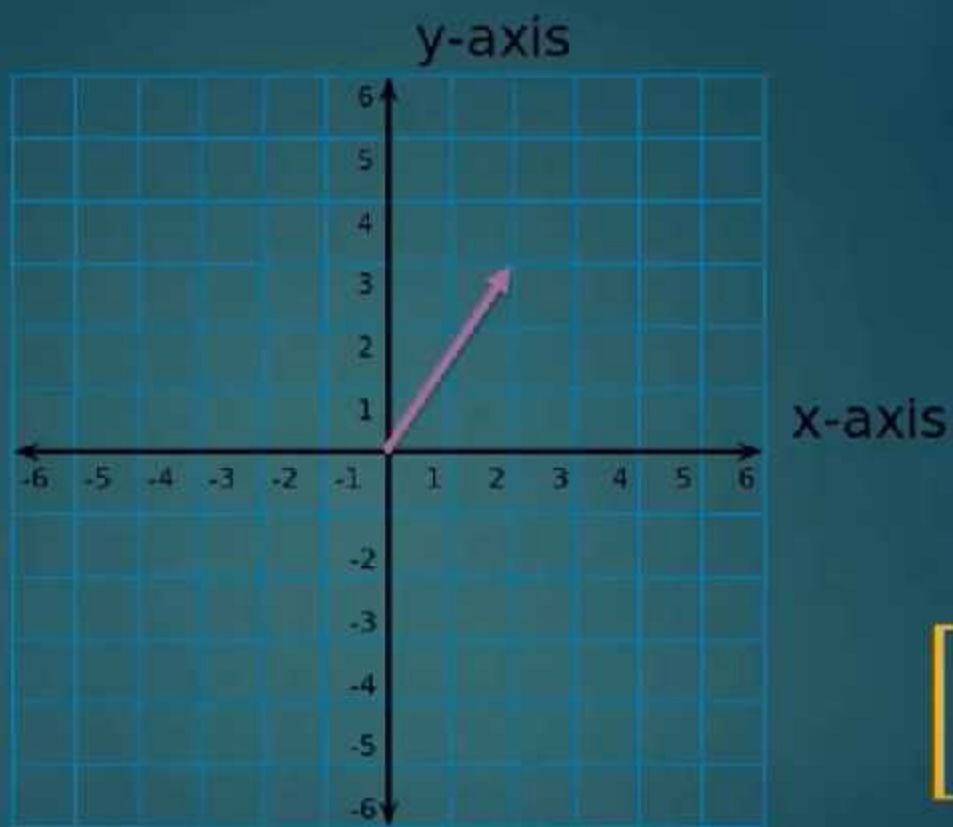


$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

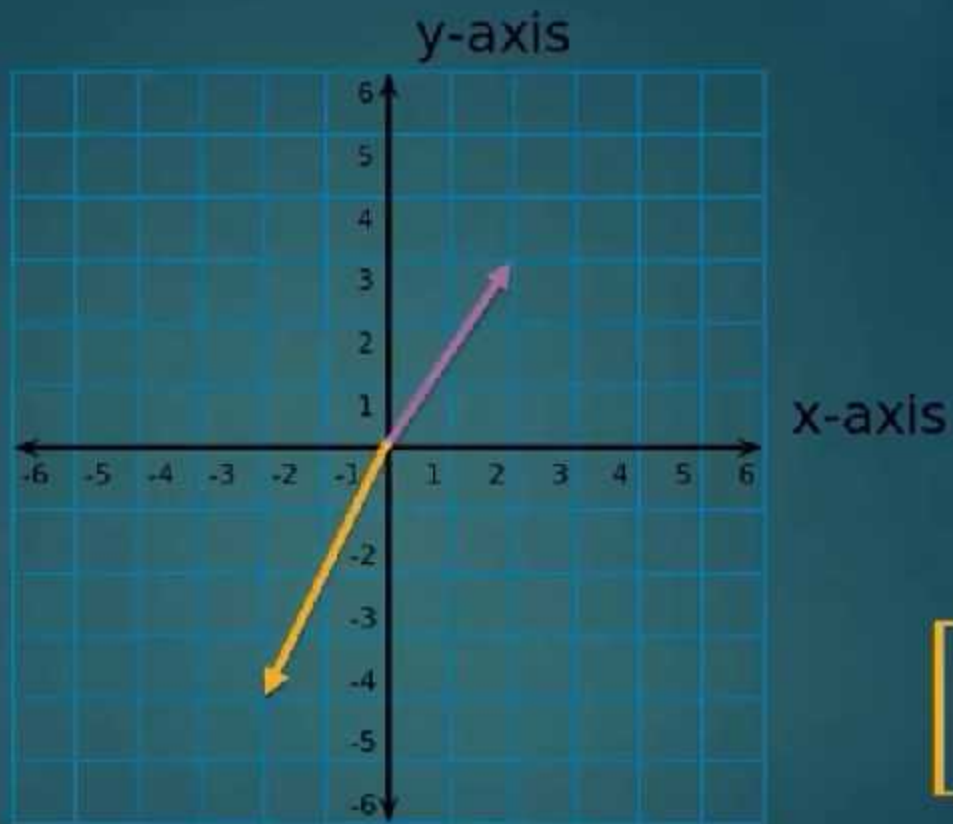


$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ -4 \end{bmatrix}$$

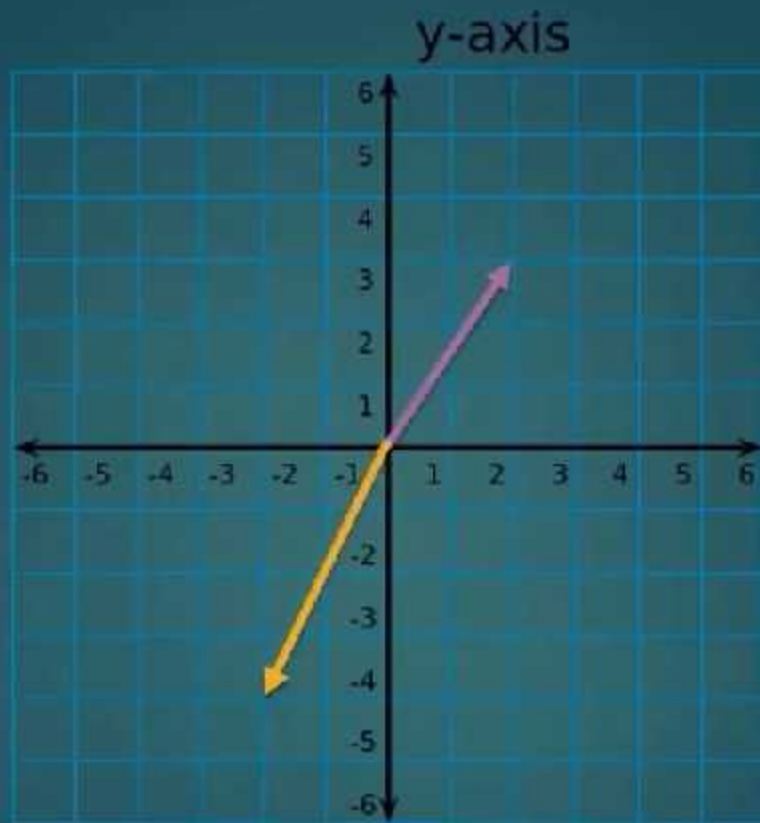


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Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ -4 \end{bmatrix}$$



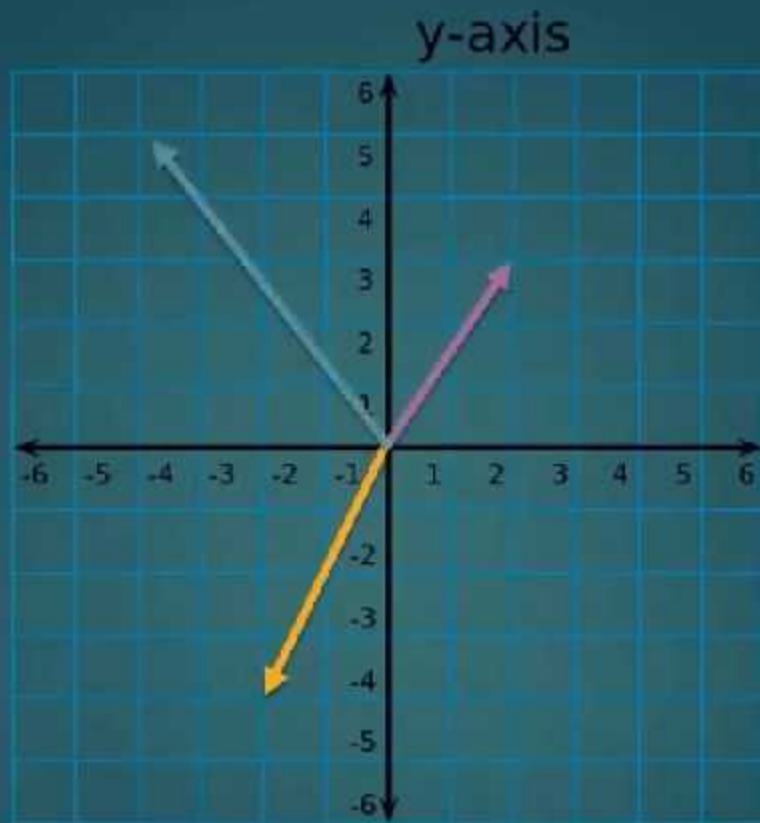
$$\begin{bmatrix} -4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ -4 \end{bmatrix}$$



$$\begin{bmatrix} -4 \\ 5 \end{bmatrix}$$

x-axis

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Operations on Vectors

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 9 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 8 \end{bmatrix}$$

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$$2 \begin{bmatrix} 9 \\ 5 \end{bmatrix} = \begin{bmatrix} 18 \\ 10 \end{bmatrix}$$

Dot Product or Inner Product



Dot Product or Inner Product



$|r|$ $|s|$ θ

$$r \cdot s = |r| |s| \cos \theta$$

Dot Product or Inner Product



$|r|$

$|s|$

θ

$$r \cdot s = |r| |s| \cos \theta$$

$$|r| = 4$$

$$|s| = 5$$

$$\theta = 60$$

Dot Product or Inner Product

 $|r|$ $|s|$ θ

$$r \cdot s = |r| |s| \cos \theta$$

$$|r| = 4$$

$$|s| = 5$$

$$\theta = 60$$

$$r \cdot s = |r| |s| \cos \theta = 4 \times 5 \times \cos 60 = 10$$

Dot Product or Inner Product

$$r = 2i + 3j$$

$$s = 4i + 5j$$

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$$|r| |s| \cos \theta$$

$$= 8(1) + 10(0) + 12(0) + 15(1) = 23$$

Dot Product or Inner Product

$$r = 2i + 3j$$

$$s = 4i + 5j$$

$$r \cdot s = (2i + 3j) \cdot (4i + 5j)$$

$$= (2i \times 4i) + (2i \times 5j) + (3j \times 4i) + (3j \times 5j)$$

$$|r| |s| \cos \theta$$

$$= 8(1) + 10(0) + 12(0) + 15(1) = 23$$

$$r \cdot s = (x_1 \cdot x_2) + (y_1 \cdot y_2) = (2 \cdot 4) + (3 \cdot 5) = 23$$

Dot Product or Inner Product

$$r = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Dot Product or Inner Product

$$r = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \qquad s = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Dot Product or Inner Product

$$r = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad s = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$r \cdot s = r^T s = \begin{bmatrix} 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \end{bmatrix} = 2 \cdot 4 + 3 \cdot 5 = 8 + 15 = 23$$

Applications of Dot Product

1- Determining the angle between two intersecting vectors

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$$\cos \theta = \frac{A_x B_x + A_y B_y}{||A|| ||B||}$$

Applications of Dot Product

2- Determine the scalar projection for one vector on another



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Applications of Dot Product

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$$P = \frac{A_x B_x + A_y B_y}{||B||}$$